If it is raining p and John does not $\neg q$
have his umbrella with him
then he will get wet. r

It is indeed raining.

John is not wet.

Therefore, John must be having his umbrella with him

If p and not q then r.

p

$\neg r$

facts

Therefore, q.

deduce

rigorous

truth-preserving

Propositional Logic

Declarative sentences

→ as propositions

The number 5 is even.

The number 7 is prime.

Today is Tuesday.

Atomic propositions

atoms

Reasoning calculus

↳ to draw conclusions from a set of assumptions.

→ in a reliable way

truth preserving

p: The number 5 is even.

q: The number 5 is prime.

negation $\neg p$: The number 5 is not even.

conjunction $p \wedge q$: The number 5 is even and the number 5 is prime.

disjunction $p \vee q$: The number 5 is even or the number 5 is prime.

implication $p \to q$: If the number 5 is even then the number is prime.

$$(p \wedge (q)) \to ((r) \vee \neg s)$$

$$p \to q \to r$$

$$p \to (q \to r)$$

$$(p \wedge q) \to ((\neg r) \vee s)$$

$$p \wedge (q \vee r)$$

$$(p \wedge q) \vee r$$

Binding priorities

$\neg$ binds the most tightly.

$\wedge$, $\vee$ bind less tightly than $\neg$ but more than $\to$

$\to$ is right associative

Natural Deduction

collection of proof rules that allows us to infer formulas from other formulas.

$$\phi_1 \qquad \phi_2 \qquad \phi_3$$

$$(p \land \neg q) \to r,\ p,\ \neg r \vdash q$$

sequent

derives

a sequent is valid if a proof for it can be found.

Rules for conjunction ($\wedge$)

$$\frac{\phi \qquad \psi}{\phi \wedge \psi} \; \wedge i$$

$$\frac{\phi \wedge \psi}{\phi} \; \wedge e_1$$

$$\frac{\phi \wedge \psi}{\psi} \; \wedge e_2$$

$$p \land q,\ r \vdash q \land r$$

1.	$p \land q$	assumption / premise
2.	r	premise
3.	q	$\land e_2$, 1
4.	$q \land r$	$\land_i$ 3, 2

$$(p \wedge q) \wedge r, \quad s \wedge t \vdash q \wedge s$$

1.	$(p \wedge q) \wedge r$	premise
2.	$s \wedge t$	premise
3.	s	$\wedge e_1$, 2
4.	$(p \wedge q)$	$\wedge e_1$, 1
5.	q	$\wedge e_2$, 4
6.	$q \wedge s$	$\wedge i$ 5, 3

Rules for double negation

$$\frac{\phi}{\neg\neg\phi}\ \neg\neg i \qquad \frac{\neg\neg\phi}{\phi}\ \neg\neg e$$

Rules for implication

$$\frac{\phi \qquad \phi \to \psi}{\psi}\ \to e$$

(Modus ponens / MP)

Modus tollens (MT)

$$\frac{\phi \to \psi \qquad \neg\psi}{\neg\phi}$$

$\phi \to q$

premise conclusion

Implication introduction

$$\frac{\boxed{\begin{array}{ll} \phi & \text{assumption} \\ \vdots & \\ \psi & \end{array}}}{\phi \to \psi} \; \to i$$

$$\phi \to \psi \vdash \neg\psi \to \neg\phi$$

1.	$\phi \to \psi$	premise	
2.	$\neg\psi$	assumption	
3.	$\neg\phi$	MT	1, 2
4.	$\neg\psi \to \neg\phi$	$\to i$	2-3

$\vdash (q \to r) \to ((\neg q \to \neg p) \to (p \to r))$

1. $q \to r$ — assumption
2. $(\neg q \to \neg p)$ — assu–
3. p — assumption
4. $\neg\neg p$ — $\neg\neg$i
5. $\neg\neg q$ — MT 2,4
6. q — $\neg\neg$e
7. r — MP 1,6
8. $(p \to r)$ — $\to$i 3-7
9. $(\neg q \to \neg p) \to (p \to r)$ — $\to$i 2-8
10. — $\to$i 1-9

$$\frac{\neg q \to \neg p \quad \neg\neg p}{\neg\neg q}$$

q

r